

Biotechnology is not a recent development. In prehistoric times, ethanol was probably the first organic substance of alcoholic fermentation. Microorganisms have been used to produce beer, vinegar, yoghurt, and cheese for over 8 millennia. Biotechnology covers a wealth of specialized disciplines, which include not only the new genetic engineering techniques but also “old fermentation”, which our ancestors practised thousands of years ago. Its origins go back to the dawn of civilization, when women first began to domesticate animals and plants gradually changing their characteristics. Selection for certain characteristics may have occurred on purpose, as well as unwittingly. Discoveries of fermentation, use of fungi for taste development and preservation led to culture selection and improvement of microorganisms for food preparation. All this is known as old biotechnology. Though the term was coined by Karl Ereky as late as 1919, biotechnology is as old as today’s man.

In the second half of the 19th century, the pioneering work of Pasteur and a number of other microbiologists and chemists showed that in microorganism, many important physiological, metabolic reactions take place, which could be considered for technical utilization. Following this realization, manufacturing processes for a whole series of primary metabolic products were rapidly developed. The major products of old biotechnology industry are food and flavour ingredients, industrial alcohol, antibiotics and citric acid, which amount about hundreds of billion dollars annually on world-wide basis. The first period of historical development of biotechnology continues to influence one present trends and is currently still being extended in parallel to other, more modern, areas of biotechnology. In the past few decades, dramatic new developments in the ability to select and manipulate genetic material have sparked unprecedented interest in the medical uses of living organisms. As farmers and industry became more interdependent and as the biotech industries developed, plants and microbes were modified to suit industrial needs. In recent years, many exciting developments have taken place, and especially in the eyes of the public, the field has gained greatly its importance.

OLD BIOTECHNOLOGY VS NEW BIOTECHNOLOGY

The origin of biotechnology can be traced back to prehistoric times, when microorganisms were used for processes like fermentation. Although a molecular biologist may consider cloning of DNA to be the most important event in the history of biotechnology, the latter has actually been rediscovered in 1970s for the third time during the present century. In 1920s, *Clostridium acetobutylicum* was used by Chaim Weizmann for converting starch into butanol and acetone; the

latter was an essential component of explosives during World War I. This raised hopes for commercial production of useful chemicals through biological processes, and may be considered the first rediscovery of biotechnology in the present century. Similarly, during World War II (in 1940s), the production of penicillin (as an antibiotic discovered by Alexander Fleming in 1929) on a large scale from cultures of *Penicillium notatum*, marked the second rediscovery of biotechnology. This was the beginning of an era of antibiotic research. The third rediscovery of biotechnology is its recent reincarnation in the form of recombinant-DNA technology, which led to the development of a variety of gene technologies and is thus considered to be the greatest scientific evolution of this century. Thus, even though biotechnology had its origin in the past, it has assumed special significance only in 1970s and 1980s. The field of modern biotechnology is thought to have largely begun on June 16, 1980, when the United States Supreme Court ruled that a genetically modified microorganism could be patented in the case of *Diamond v. Chakrabarty*. Indian-born Ananda Chakrabarty, working for General Electric, had developed a bacterium (derived from the *Pseudomonas* genus) capable of breaking down crude oil, which he proposed to use in treating oil spills.

Biotechnology, as the word indicates, is the product of interaction between the science of biology and technology. This relationship between science and technology has been observed to be complex, so that not only the science has influenced technology, but the technology has also influenced science. Because of this complex relationship and its major impact on human welfare, it is believed that biotechnology in future may become a major force for human existence. Already, the products of biotechnology (including diagnosis, prevention and cure of diseases; new and cheaper chemical products, e.g., pharmaceutical drugs and the new food sources; devices for environment protection and energy conservation, etc.) are playing a very important role in employment, productivity, trade, economies and the quality of human life throughout the world. It has been recognized that a complete definition of biotechnology is difficult due to such a wide range of its usage. Fermentation, by some microorganisms, formation of yoghurt (curd) and cheese from milk and that of vinegar from molasses, production of antibiotics like penicillin from certain fungi and the processes of baking and brewing are often included in describing what is called old biotechnology. The other examples of reaction (PCR), cell culture and fusion and bio-processing which became possible only through the researches in molecular biology have been described as new biotechnology (according to OTA, i.e., Office of Technology Assessment).

WHAT IS BIOTECHNOLOGY?

At its simplest, biotechnology is the commercial and large scale exploitation of living organisms or their components, e.g., enzymes for the welfare of mankind through molecular genetics. The major foundation of biotechnology arose out of empirical developments in the field of molecular genetics. Understanding the molecular basis of life processes and life will help in the development of molecular biology and in turn biotechnology.

Year	Historical Events in Biotechnology
1750 BC	The Sumerians brew beer.
500	The Chinese use moldy soybean curds as an antibiotic to treat boils.
250	The Greeks use moldy crop rotation to maximize soil fertility.
100	Powdered Chrysanthemum is used in China as an insecticide.
1590 AD	The first compound microscope is invented by Zacharias Jansen.
1663	Cells are first described by Hooke.
1675	Leeuwenhoek discovers protozoa and bacteria.
1797	Jenner inoculates a child with a viral vaccine to protect him from smallpox.
1833	The cell nucleus is discovered. The first enzymes are isolated.
1855	The <i>Escherichia coli</i> bacterium is discovered as production tool for biotechnology.
1863	Mendel, in his study of peas, discovers that traits were transmitted from parents to progeny by discrete, independent units, later called genes.
1869	F. Miescher discovers Nuclein (now known as DNA) from pus cells.
1877	A technique for staining and identifying bacteria is developed by Koch.
1879	Flemming discovers chromatin, the rod-like structures inside the cell nucleus.
1883	The first rabies vaccine is developed.
1907	The first <i>in vivo</i> culture of animal cells is reported.
1909	Genes are linked with hereditary disorders.
1911	The first cancer-causing virus is discovered by Rous.
1915	Phages, or bacterial viruses are discovered.
1919	The word "biotechnology" is first used by a Hungarian agricultural engineer.
1927	Muller discovers that X-rays cause mutation.
1928	Fleming discovers penicillin, the first antibiotic.
1941	The term "genetic engineering" is first used by a Danish microbiologist.
1942	The electron microscope is used to identify and characterize a bacteriophage- a virus that infects bacteria.
1943	Oswald T. Avery demonstrates that DNA is the "transforming factor" and is the material of genes.
1951	McClintock discovers transposable elements, or "jumping genes," in corn.
1953	Watson and Crick reveal the three-dimensional structure of DNA.
1954	Cell-culturing techniques are developed.
1956	Kornberg discovers the enzyme DNA polymerase I, leading to an understanding of how DNA is replicated.
1960	Exploiting base pairing, hybrid DNA-RNA molecules are created. Messenger RNA is discovered.
1961	The genetic code is understood for the first time.
1967	The first automatic protein sequencer is perfected.
1969	An enzyme is synthesized <i>in vitro</i> for the first time.
1970	Specific restriction nucleases are identified, opening the way for gene cloning reverse transcriptase (RT) is discovered independently in murine and avian retroviruses.
1971	RT is shown to have ribonuclease H (Rnase H) activity.
1972	Purified RT is first used to synthesize cDNA from purified mRNA <i>in vitro</i> .
1973	Cohen and Boyer perform the first successful recombinant DNA experiment, using bacterial genes.
1974	The NIH forms a Recombinant DNA Advisory Committee to oversee recombinant genetic research.
1975	Colony hybridization and Southern blotting are developed for detecting specific DNA sequences. The first monoclonal antibodies are produced.
1976	The tools of recombinant DNA are first applied to a human inherited disorder.
1977	Genetically engineered bacteria are used to synthesize human growth protein.

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4 Biotechnology in Medicine and Agriculture

1978	Hutchinson and Edgell show that it is possible to introduce specific mutations at specific sites in a DNA molecule.
1979	The first monoclonal antibodies are produced.
1980	The U.S. Supreme Court, in a landmark case <i>Diamond v. Chakrabarty</i> , approves the patenting of genetically engineered life forms. The U.S. patent for gene cloning is awarded to Cohen and Boyer.
1981	The first 'gene-synthesizing' machines are developed. The first genetically engineered plant is reported.
1982	Humulin, Genentech's human insulin drug produced by genetically engineered bacteria for the treatment of diabetes, is the first biotech drug to be approved by the Food and Drug Administration.
1983	The Polymerase Chain Reaction (PCR) technique is conceived. The first genetic transformation of plant cells by Ti plasmids is performed. The first artificial chromosome is synthesized.
1984	The DNA fingerprinting technique is developed. The first genetically engineered vaccine is developed. Chiron clones and sequences the entire genome of the HIV virus.
1985	Fully active murine RT is cloned and over-expressed in <i>E. coli</i> .
1986	The first biotech-derived interferon drugs for the treatment of cancer, Biogen's Intron A and Genentech's Roferon A, are approved by the FDA. The first genetically engineered human vaccine, Chiron's Recombivax HB, is approved for the prevention of hepatitis B.
1987	Advanced Genetic Sciences's Frostban, a genetically altered bacterium that inhibits frost formation on crop plants, is field tested on strawberry and potato plants in California, the first authorized outdoor tests of an engineered bacterium. Reverse transcription and PCR are combined to amplify mRNA sequences.
1988	Congress funds the Human Genome Project, a massive effort to map and sequence the human genetic code as well as the genomes of other species.
1989	Microorganisms are used to clean up the Exxon Valdez oil spill.
1990	The first federally approved gene therapy treatment is performed successfully on a 4-year old girl suffering from an immune disorder.
1991	Amgen develops Neupogen, the first of a new class of drugs called colony stimulating factors, for the treatment of low white blood cells in chemotherapy patients. Immunex's Leukine, used to replenish white blood counts after bone marrow transplants, is approved.
1992	The three-dimensional structure of HIV RT is elucidated. Recombinate, developed by Genetics Institute and used in the treatment of hemophilia A, becomes the first genetically engineered blood clotting factor approved. Chiron's Proleukin is approved for the treatment of renal cell cancer.
1993	The FDA declares that genetically engineered foods are "not inherently dangerous" and do not require special regulation.
1994	Calgene's Flavr Savr tomato, engineered to resist rotting, is approved for sale.
1995	The first full gene sequence of a living organism other than a virus is completed for the bacterium " <i>Haemophilus influenzae</i> ."
1996	Biogen's Avonex is approved for the treatment of multiple sclerosis. The company builds a \$50 million plant in Research Triangle Park, N.C., to manufacture the recombinant interferon drug.
1997	Scottish scientists report cloning a sheep (Dolly), using DNA from adult sheep cells. A new DNA technique combines PCR, DNA chips, and a computer program, providing a new tool in the search for disease-causing genes.
1998	Human skin is produced <i>in vitro</i> . Embryonic stem cells are used to regenerate tissue and create disorders mimicking diseases. The first complete animal genome for the elegans worm is sequenced.
1999	The complete genetic code of the human chromosome is first deciphered. The rising tide of public opinion in Europe brings biotech food into the spotlight.
2000	"Golden Rice," modified to make vitamin A, promises to help third-world countries alleviate night blindness.

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2001	The sequence of the human genome is published in Science and Nature, making it possible for researchers all over the world to begin developing treatments.
2002	Kurt Wüthrich developed nuclear magnetic resonance spectroscopy for determining the three-dimensional structure of biological macromolecules in solution.
2003	Dolly, the cloned sheep that made headlines in 1997, is euthanized after developing progressive lung disease. Dolly was the first successful clone of a mammal.
2004	Approval of Xolair (omalizumab, for use in asthma); Raptiva (efalizumab, for use in psoriasis); Avastin (bevacizumab, for the treatment of cancer).
2005	A first draft of the chimpanzee genome is published.
2006	Discovery of RNA interference-gene silencing by double-stranded RNA.
2007	Discovery of principles for introducing specific gene modifications in mice by the use of embryonic stem cells.
2008	Paolo Macchiarini carries out the world's first tissue-engineered whole organ transplant, replacing a Colombian woman's windpipe made with the patient's own stem cells.
2009	Discovery of how chromosomes are protected by telomeres and the enzyme telomerase.
2010	Dr. Craig Venter and his team of geneticists at the J. Craig Venter Institute (JCVI) create a synthetic life form by using chemically-created chromosomes in empty cells, causing those cells to multiply.
2011	Discovery and development in nanoparticles based products.
2012	Development of genetically engineered stem cells to seek out and kill HIV in mice.

Definitions

Biotechnology, broadly defined, includes any technique that uses living organisms (or parts of organism) to make or modify products, to improve plants or animals, or to develop microorganisms for specific uses. However, other definitions are also given below:

- Biotechnology is the application of biological organisms, system or process to manufacturing and service industries.
- Biotechnology is the integrated use of biochemistry, microbiology, and engineering sciences in order to achieve technological (industrial) application of the capabilities of microorganisms, cultured tissue cells and parts thereof.
- Biotechnology is a technology using biological phenomena for copying and manufacturing various kinds of useful substances.
- Biotechnology is the controlled use of biological agents, such as microorganisms or cellular components, for beneficial use.
- Any technological application that uses biological systems, living organisms, or derivatives thereof, to make or modify products or processes for specific use.

Developments in Biotechnology

Biotechnology is the use of microbiology, biochemistry, genetics and engineering in an integrated fashion with the goal of using microorganisms and cell and tissue cultures (or their parts) to manufacture useful products. Biotechnology is developing by close interweaving between these areas and can develop further only by means of an interdisciplinary cooperation between them. Basic research has produced a large number of results in the field of microbiology, biochemistry, enzyme research, molecular biology, genetic engineering and, chemical engineering, which are

being practically used in biotechnological applications. Biotechnology is therefore an application oriented science of microbiology and biochemistry that is closely connected with technical chemistry and chemical engineering. The field of operation of biotechnology consists of three large areas of life science (a) microbiology, including microbial genetics, (b) biochemistry including physical chemistry and organic chemistry, and (c) process engineering and apparatus constructions.

With the discovery of restriction enzymes in 1970, biotechnology was given a boost that opened the avenue of genetic engineering. The freedom to play around with DNA molecules resulted in dramatic use of gene cloning and gene transfer technology (Modern Biotechnology). Modern biotechnology is only 50 years old, and in the last decades it has been witnessing tremendous developments. With the development of new approaches and modern techniques, traditional biotechnology industries are also acquiring new horizons enabling them to improve the quality of their products and increase the productivity of their systems. In the past few decades, dramatic new developments in the ability to select and manipulate genetic material have sparked unprecedented interest in the uses of living organisms.

In the 1980s, biotechnology became a major growth area. Many products can be manufactured only biotechnologically. This applies particularly to most of the secondary metabolic products e.g., antibiotics, etc. Since last thousands of years, biological processes have central importance in food industry, but it is only in the last 100 years that they have been applied more intensively in chemical technology. At present, modern biotechnology is capable of becoming a part of not only medical industry but also other industrial developments of increasing importance like agro-industries. It is predicted that modern biotechnology will account for a much larger fraction of the total biotechnology industry in years to come.

Biotechnology has applications in four major industrial areas, including health care (medical), crop production and agriculture, non-food (industrial) uses of crops and other products (e.g. biodegradable plastics, vegetable oil, biofuels), and environmental uses. For example, one application of biotechnology is the directed use of organisms for the manufacture of organic products (examples include beer and milk products). Another example is using naturally present bacteria by the mining industry in bioleaching. Biotechnology is also used to recycle, treat waste, clean up sites contaminated by industrial activities (bioremediation), and also to produce biological weapons.

A series of derived terms have been coined to identify several branches of biotechnology, for example:

- **Bioinformatics** is an interdisciplinary field which addresses biological problems using computational techniques, and makes the rapid organization and analysis of biological data possible. The field may also be referred to as *computational biology*, and can be defined as, “conceptualizing biology in terms of molecules and then applying informatics techniques to understand and organize the information associated with these molecules, on a large scale.” Bioinformatics plays a key role in various areas, such as functional genomics, structural genomics, and proteomics, and forms a key component in the biotechnology and pharmaceutical sector.
- **Blue biotechnology** is a term that has been used to describe the marine and aquatic applications of biotechnology, but its use is relatively rare.

- **Green biotechnology** is biotechnology applied to agricultural processes. An example would be the selection and domestication of plants via micropropagation. Another example is the designing of transgenic plants to grow under specific environments in the presence (or absence) of chemicals. One hope is that green biotechnology might produce more environment-friendly solutions than traditional industrial agriculture. An example of this is the engineering of a plant to express a pesticide, thereby ending the need of external application of pesticides. An example of this would be Bt corn. Whether or not green biotechnology products such as this are ultimately more environment friendly is a topic of considerable debate.
- **Red biotechnology** is applied to medical processes. Some examples are the designing of organisms to produce antibiotics, and the engineering of genetic cures through genetic manipulation.
- **White biotechnology**, also known as industrial biotechnology, is biotechnology applied to industrial processes. An example is the designing of an organism to produce a useful chemical. Another example is the using of enzymes as industrial catalysts to either produce valuable chemicals or destroy hazardous/polluting chemicals. White biotechnology tends to consume less in resources than traditional processes used to produce industrial goods. The investment and economic output of all of these types of applied biotechnologies is termed as bio-economy.

COMPONENTS OF BIOTECHNOLOGY

To keep pace with the ever expanding knowledge in the field of molecular genetics and molecular biology it necessitates harnessing this knowledge for the development of potential technologies and products for the welfare of mankind. In order to facilitate the development of biotechnology, we have to look upon three major components: objective, tools and methodology or techniques. Thus, in biotechnology, objective is the product of gene or metabolism; the tools are microorganisms (bacteria, fungi and viruses) and cultured cells of higher organisms; and the techniques employed are those used for development of commercially important biotechnologies.

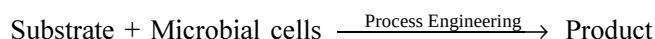
Biological Organisms as Tools

Biotechnology always deals with reactions which, in principle, are of biological nature. These reactions are performed either by living microbial cells or plant and animal cells and their tissues or by enzymes from cells or parts of cells. The microorganisms, such as bacteria, actinomycetes, cyanobacteria, and yeasts and other fungi, viruses and mycoplasmas widely occur in nature. They function in industry as synthesizers of several useful products of fermentation and bioconversion. They have been the major tools in the study of molecular genetics and molecular biology which, in turn biotechnology. The production of biomass from the organisms or parts of organisms mentioned is also an area of biotechnology. In 1960s, considerable excitement was generated by the prospect of using microbial cells (or biomass) as a source of protein, the so-called single cell protein (SCP). However, introduction of SCP has not been a success.

Microorganisms figure principally as tools in biotechnology, especially for genetic engineering and gene cloning studies.

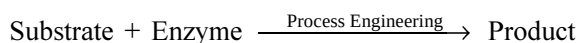
Biotechnology Processes

Microorganisms are the deciding factor of a biotechnological process. Basically, most biotechnological processes can be reduced to a simple equation:



Process engineering consists of the manufacturing process proper and the subsequent recovery procedure (isolation of the product or the cells).

The product may be cell biomass, a metabolic product of the cells, or a transformation product of a given starting material, which is formed by the living microorganism or enzymes. Furthermore, an initially complex substrate can be replaced by a definite chemical compound, which is converted into a product in one or a few steps. Then the equation runs, for example:



These reactions are already simplified biotechnological processes. In general, however, biotechnological processes are altogether more complex and employ various technologies.

The Technologies

The novel methodology and techniques used in biotechnology are extremely powerful because they allow a large amount of control over biological system. Biotechnology as practised today is much more than recombinant DNA technology. It includes various technologies (as discussed below). However, if these are performers, gene manipulation is the star of show. The ability to splice genes has revolutionized the industry and is leading to the development of countless new products and improved methods for well established processes, which have wide applications in agriculture and medicine. However, besides recombinant DNA technology, some other potential biotechnologies have wide applications in agricultural and medical sciences and are briefly described below. (Principles and practices of these potential biotechnologies in agriculture and medicine are covered in the form of full chapters subsequently in this book).

Recombinant DNA technology

Recombinant DNA technology is a set of molecular techniques for locating, isolating, altering, and studying DNA segments. The term *recombinant* is used because frequently the goal is to combine DNA from two distinct sources. Genes from two different bacteria might be joined, for example, or a human gene might be inserted into a viral chromosome. Commonly called genetic engineering, recombinant DNA technology now encompasses an array of molecular techniques that can be used to analyze, alter, and recombine virtually any DNA sequences. In addition to providing valuable new information about the nature and function of genes, recombinant DNA technology has many applications. These applications include the production of pharmaceuticals and other chemicals,

specialized bacteria, agriculturally important plants, and genetically engineered farm animals. The technology is also used extensively in medical testing and, in a few cases, is even being used to correct human genetic defects. Hundreds of firms now specialize in developing products through genetic engineering, and many large multinational corporations have invested enormous sums of money in recombinant DNA research. These techniques are also used in many other fields, including biochemistry, microbiology, developmental biology, neurobiology, evolution, and ecology. Recombinant DNA technology is also used to create a number of commercial products, including drugs, hormones, enzymes, and crops. An entirely new industry—biotechnology—has grown up around the use of these techniques to develop new products. In medicine, recombinant DNA techniques are used to probe the nature of cancer, diagnose genetic and infectious diseases, produce drugs, and treat hereditary disorders.

Nucleic acid probes technology

Nucleic acid probes technology has been developed that relies on the use of DNA or RNA probes to detect particular gene sequences. The application of the technique includes the diagnosis of genetic and infectious diseases. Through intensive research efforts in both chemical synthesis and bioassay development, molecular probes scientists have developed rapid and exceptionally sensitive fluorescence-based assays for quantitation of nucleic acids (DNA, RNA) and protein samples. The double-helical structure of nucleic acids, in which one strand binds specifically to its exact complement, is a fortuitous design for the study of biology. The ease of constructing assays based on this feature is responsible for the incredibly fast pace that has characterized molecular biology research since its inception. With the incorporation of fluorescence technologies, the ability to design multiplexed and high-throughput assays has increased the pace still further to the point in which sophisticated bioinformatics is required to analyze the huge outpouring of data. Experiments in which labeled sample nucleic acids are hybridized to nucleic acids on solid supports, most notably Southern and Northern blots, have been core technologies in the field of molecular biology. Conventionally, samples were labeled with radioisotopes, hybridized to nucleic acids on the membrane and imaged for long periods with X-ray film. Although some users still employ radioactive labeling for microarray experiments, fluorescent labeling and detection now dominate the field. Fluorescence provides an important advantage because the majority of microarray experiments rely on comparison of hybridization signals between two or more samples, and fluorescence detection allows the analysis of multiple samples on the same array, minimizing artifact variation in signals.

Polymerase chain reaction technology

The polymerase chain reaction (PCR) is a biochemistry and molecular biology technique originally conceived by Kary Mullis for exponentially amplifying DNA via enzymatic replication without using a living organism (such as *E. coli* or yeast). As PCR is an *in vitro* technique, it can be performed without restrictions on the form of DNA, and it can be extensively modified to perform a wide array of genetic manipulations. PCR is commonly used in medical and biological research labs for a variety of tasks, such as the detection of hereditary diseases, the identification of genetic fingerprints, the diagnosis of infectious diseases, the cloning of genes, paternity testing, and DNA

computing. PCR was invented by Kary Mullis. At the time he thought up PCR in 1983, Mullis was working in Emeryville, California for Cetus Corporation, one of the first biotechnology companies. There, he was charged with making short chains of DNA for other scientists. Mullis has written that he conceived of PCR while cruising along the Pacific Coast Highway one night in his car. He was playing in his mind with a new way of analyzing changes (mutations) in DNA when he realized that he had instead invented a method of amplifying any DNA region. Mullis has said that before his trip was over, he was already savoring the prospects of a Nobel Prize. He shared the Nobel Prize in Chemistry with Michael Smith in 1993. PCR technology has wide application such as genetic fingerprinting, paternity testing, detection of hereditary diseases, cloning genes, mutagenesis, analysis of ancient DNA, genotyping of specific mutations, comparison of gene expression and many more.

DNA-protein interaction technology

DNA–protein interactions are fundamental to the existence of life forms, providing the key to the genetic plan as well as mechanisms for its maintenance and evolution. The specific binding of a protein to a nucleic acid is a first step in several central processes in a living cell. Sequence-specific protein–DNA interactions are crucial for the functional read-out of genetic information. The study of these interactions is therefore fundamental to our understanding of growth, development, differentiation, evolution, and disease. The manipulation of DNA-protein interactions is also becoming increasingly important to the biotechnology industry, permitting among other things the reprogramming of gene expression. DNA-protein complexes may be studied at three distinct levels—at the level of the DNA, of the protein, and of the complex. At the level of the DNA, the DNA binding site may be delimited and exact base sequence requirements defined. The DNA conformation can be studied and the exact bases contacted by the protein identified. At the protein level, the protein species binding a given DNA sequence can be identified. The amino acids contacting DNA and the protein surface facing the DNA may be defined and the amino acids essential to the recognition process can be identified. Furthermore, the protein's tertiary structure and its conformational changes on complex formation can be studied. Finally, global parameters of a DNA-protein complex such as stoichiometry, the kinetics of its formation and dissociation, its stability, and the energy of interaction can be measured. Filter binding, electrophoretic mobility shift assay (EMSA/gel shift), DNaseI footprinting, and Southwestern blotting have been the most commonly used techniques to identify potentially interesting DNA target sites and to define the proteins that bind them. DNA-protein crosslinking has in recent years become a very important approach to investigate the relative positions of proteins in multicomponent protein-DNA complexes such as the transcription initiation complex. When combined these methods are capable of providing a picture of the protein-DNA complex with a great deal of molecular detail, surpassed only by the more demanding methods of crystallography and nuclear magnetic resonance (NMR).

Genomics technology

Genomics is the study of the genomes of organisms. The field includes intensive efforts to determine the entire DNA sequence of organisms and fine-scale genetic mapping efforts. The field

also includes studies of intragenomic phenomena such as heterosis, epistasis, pleiotropy and other interactions between loci and alleles within the genome. In contrast, the investigation of single genes, their functions and roles, something very common in today's medical and biological research, and a primary focus of molecular biology, does not fall into the definition of genomics unless the aim of this genetic pathway, and functional information analysis is to elucidate its effect on, place in, and response to the entire genome's networks. For the United States Environmental Protection Agency, "the term "genomics" encompasses a broader scope of scientific inquiry associated technologies than when genomics was initially considered. A genome is the sum total of all an individual organism's genes. Thus, genomics is the study of all the genes of a cell, or tissue, at the DNA (genotype), mRNA (transcriptome), or protein (proteome) levels. A major branch of genomics is still concerned with sequencing the genomes of various organisms, but the knowledge of full genomes has created the possibility for the field of functional genomics, mainly concerned with patterns of gene expression during various conditions.

Proteomics technology

Proteomics technology is an interdisciplinary science that includes biology and protein chemistry. A broader definition of proteomics includes systematic experimental and computational attempts at determining protein-protein interaction maps for an entire organism. This information is crucial to an understanding of how genes work in concert to generate a working cell. Functional proteomics constitutes an emerging research area in the proteomic field whose approaches are addressed towards two major targets: the elucidation of the biological function of unknown proteins and the definition of cellular mechanisms at the molecular level. The identification of interacting proteins in stable complexes *in vivo* is essentially achieved by affinity-based procedures. The basic idea is to express the protein of interest with a suitable tag to be used as a bait to fish its specific partners out from a cellular extract. Individual components within the multi-protein complex can then be identified by mass spectrometric methodologies. The association of an unknown protein with partners belonging to a specific protein complex involved in a particular mechanism is strongly suggestive of the biological function of the protein. Furthermore, a detailed description of the cellular signaling pathways might greatly benefit from the elucidation of protein-protein interactions *in vivo*. Since proteomics is a young and rapidly evolving field, the best approach is to gain a general understanding of the questions and technologies involved and then pursue the primary literature for detailed information on the latest developments.

Signalomics technology

The goal of signalomics technology is to track down faulty cellular signal proteins and to develop custom-made therapeutics, theranostics and diagnostics for complex illnesses that result from errors in cellular signal transduction. The ability to specifically influence cellular signals will make it possible to create individualized medications which are custom-designed for the patient. This is the revolutionary strategy that signalomics is pursuing. In its study of intracellular processes, signalomics makes use of functional proteomics, nano-biotechnology, computational biology and bioanalysis using fluorescent dyes. Signalomics makes it possible to observe dynamic molecular

processes and interactions even in living cells. Furthermore, it is also possible to detect a variety of binding partners simultaneously.

Metabolomics technology

Metabolomics is the “systematic study of the unique chemical fingerprints that specific cellular processes leave behind”—specifically, the study of their small-molecule metabolite profiles. The metabolome represents the collection of all metabolites in a biological organism, which are the end products of its gene expression. Thus, while mRNA gene expression data and proteomic analyses do not tell the whole story of what might be happening in a cell, metabolic profiling can give an instantaneous ‘snapshot’ of the physiology of that cell. One of the challenges of systems biology is to integrate proteomics, transcriptomics, and metabolomics information to give a more complete picture of living organisms. Metabolomics was conceived in 1970 by Arthur Robinson, an assistant professor in the Department of Chemistry at the University of California, San Diego. Metabolomics can be an excellent tool for determining the phenotype caused by a genetic manipulation, such as gene deletion or insertion. Metabolic profiling (especially of urine or blood plasma samples) can also be used to detect the physiological changes caused by toxic insult of a chemical (or mixture of chemicals).

Bioinformatics and *in silico* technology

Bioinformatics and *in silico* technology occupies a unique niche amongst the sciences, lying at the intersection of biology, genetics, biochemistry, computer science, mathematics, statistics, and numerous other allied fields. The inherent strength of the field of bioinformatics comes from the relationships between investigators in these allied fields. Collaborations between these individuals has led to the development of novel methods and approaches, furthering advances in each of these areas. Such collaborations also set the stage for the piloting of experiments on computers, followed by the verification of the computational results in the laboratory. DNA sequences generated by the human genome project, protein structures, gene expression patterns, etc. carry with them an enormous amount of valuable information. To uncover all the information, however, will still take many more years of research. Bioinformatics, as we have seen, is set to become an indispensable tool. This is not to say that wet-bench work is not required; on the contrary, bioinformatics will give researchers the tools necessary to guide their bench work and handle computers as easily as they do the microscope. *In silico* technologies in drug target identification and validation addresses the challenge of testing a growing number of new potential targets and reviews currently available *in silico* approaches for identifying and validating these targets and determine the biological functionality of potential targets more effectively.

System biology technology

Systems biology aims at understanding biological systems at system level. Systems biology, just like all other emerging scientific disciplines, is built on multiple efforts that share the vision. Systems biology clearly includes both experimental and computational or analytical studies. However, systems biology is not a mere combination of molecular biology and computing science to reverse

engineer gene networks. Of course, it is one of the topics included, but system-level understanding requires more than understanding structures. Understanding of (1) system structures, (2) dynamics, (3) control methods, and (4) design methods are major four milestones of systems biology research. To understand biology at the system level, we must examine the structure and dynamics of cellular and organismal function, rather than the characteristics of isolated parts of a cell or organism. However, many breakthroughs in experimental devices, advanced software, and analytical methods are required before the achievements of systems biology can live up to their much-touted potential. Systems biology works towards understanding the intricacies of cellular life through the collaborative efforts of biologists, chemists, mathematicians and computer scientists and recently, a number of laboratories around the world have embarked upon such research agendas. Systems biology will bring about a revolution in the practice of medicine. Knowledge of specific genetic traits, coupled with multi-parameter diagnostics, will generate routine procedures for assessing health and disease status. Knowledge of disease-perturbed networks will facilitate drug discovery, and pharmacological intervention will focus on preventing disease-mediated transitions. This predictive and preventative medicine will lead naturally to a personalized medicine, in which therapeutic strategies will be tailored to individual needs. In this sense, systems biology will fundamentally transform society.

Microarray technology

Microarray is an advanced and high throughput method based on DNA chips, which consists of very high density arrays of short oligonucleotides synthesized on a silicon wafer by a combination of conventional oligonucleotide synthesis chemistry and photolithography. There are two parallel methods to monitor the genes, which are differentially expressed in a particular sample, oligonucleotide microarrays and cDNA microarrays. Both of these methods offer tremendous advantages in speed of analysis, allowing thousands of cDNA sequences to be surveyed in a single set of experiments. This method has been applied for profiling of differentially expressed genes from various plant samples and recently there are also some reports for their usefulness in profiling of ripening related genes. This technology has wide applications, such as in quantitative gene expression; mutation detection; disease profiling (diagnostics); patient monitoring (prognostics) and many more.

Antisense technology

Antisense technologies are a suite of techniques that, together, form a very powerful weapon for studying gene function (functional genomics) and discovering new and more specific treatments of diseases in humans, animals, and plants (antisense therapeutics). A conventional definition of antisense refers to the laboratory manipulation and/or modification of DNA or RNA so that its components (nucleotides) form a complementary copy of normal, or “senses”, messenger RNA (mRNA). The binding, or hybridization of antisense nucleic acid sequences to a specific mRNA target will, through a number of different mechanisms, interrupt normal cellular processing of the genetic message of a gene. This interruption, sometimes referred to as “knock-down” or “knock-out” depending upon whether or not the message is either partially or completely eliminated, allows

researchers to determine the function of that gene. The impact of biotechnology on antisense technology is expected to increase dramatically as the links between genetics, protein production and disease are better understood. Currently, antisense technology is used to design therapeutic compounds, which target specific mRNA sequences to obstruct the production of certain disease-causing proteins. Traditional drug therapies focus on a drug's interaction with the disease-causing proteins. However, antisense drug therapies inhibit the production of the disease-causing proteins altogether. Applications of antisense technology in therapeutics are expected to increase substantially as the link between diseases and genes are made.

RNA interference technology

RNA interference (RNAi) technology has become one of the key methods used in functional genomics. RNAi is used to block the expression of genes and create phenotypes that can potentially yield clues about the function of these genes. It was discovered unexpectedly during attempts to experimentally manipulate the expression of specific genes. This naturally occurring phenomenon was first described for the nematode worm *Caenorhabditis elegans*. RNAi is believed to be an ancient cellular defense against certain viruses and mobile genetic elements in both plants and animals. RNAi is initiated by long double-stranded RNA molecules, which are processed into 21-23 nucleotides long RNAs by the dicer enzyme. These small interfering RNAs (siRNAs) are then incorporated into the RNA-induced silencing complex (RISC), a protein AERNA complex, and guide a nuclease, which degrades the target RNA. RNA interference has been used for applications in biotechnology, particularly in the engineering of food plants that produce lower levels of natural plant toxins. Recently, this technology has been applied for manipulation of ripening related gene expression to improve the post-harvest life of fruits.

Hybridoma technology

Hybridoma technology for production of monoclonal antibodies was discovered in 1975 by two scientists, Georges Kohler of West Germany and Cesar Milstein of Argentina (now working in U.K.), who jointly with Niels Jerne of Denmark (now working in Germany) were awarded the 1984 Nobel Prize for Physiology and Medicine. A hybridoma, which can be considered a hybrid cell, is produced by the injection of a specific antigen into a mouse, procuring the antibody-producing cell from the mouse's spleen and the subsequent fusion of this cell with a cancerous immune cell called myeloma cell. Monoclonal antibodies (Mabs) can be produced from these hybrid cells (hybridoma cells) through a technique now popularly known as hybridoma technology. Mabs can potentially be used for many purposes including the diagnosis and treatment of diseases and the purification of proteins. The fused hybrid cells (called hybridomas), being cancer cells, will multiply rapidly and indefinitely and will produce large amounts of the desired antibodies. They have to be selected and subsequently cloned by limiting dilution. Supplemental media containing Interleukin-6 (such as briclone) are essential for this step.

Molecular diagnostics technology

Molecular technology that has been implemented into diagnostic testing is called molecular diagnostic technology. The molecular diagnostics involves the molecules-molecules interaction based on their complementarities and affinity. These include broadly ligand-binder such as the DNA-DNA, DNA-RNA, DNA-protein, protein-protein and more specifically the antigen-antibody interaction that makes the basis of development of immuno-technologies for diagnostic and analytical purposes in medical, veterinary and even in agricultural sciences. Recently, several programs that educate clinical laboratory professionals have had to incorporate molecular-based diagnostic testing into their curricula just as rapidly despite the lack of formal resources. This technology has wide applications such as detection and identification of microorganisms, DNA polymorphisms and *human identification*, molecular detection of inherited diseases, gene mutations, resolution and detection of nucleic acids, nucleic acid amplification, analysis and characterization of nucleic acids and proteins, chromosomal structure and chromosomal mutations, quality assurance and quality control in the molecular laboratory, DNA-based tissue typing, molecular oncology and many more. Molecular and cytogenetic analyses are a critical component of diagnostic testing, especially for diseases that arise from known genetic events. The identification of a molecular or chromosomal abnormality is a direct observation of the source of some diseases. Genetic lesions in single-gene disorders are detected by a variety of hybridization/amplification methods and nucleotide sequencing.

Biosensor technology

Biosensor technology deals with the use of an analytical device that converts a biological response into an electrical signal. The term 'biosensor' is often used to cover sensor devices used in order to determine the concentration of substances and other parameters of biological interest even where they do not utilize a biological system directly. Enzymes are the biologically responsive material, but it should be recognized that other biological systems may be utilized by biosensors, for example, whole cell metabolism, ligand binding and the antibody-antigen reaction. Research and development in biosensor is wide and multidisciplinary, spanning biochemistry, bioreactor science, physical chemistry, electrochemistry, electronics and software engineering. Most of this current endeavor concerns potentiometric and amperometric biosensors and colorimetric paper enzyme strips. However, all the main transducer types are likely to be thoroughly examined for use in biosensors over the next few years. A successful biosensor must be highly specific for the purpose of the analyses; stable under normal storage conditions and show good stability over a large number of assays; tiny and biocompatible; having no toxic or antigenic effects; the response should be accurate, precise, reproducible and linear over the useful analytical range; free from electrical noise; should not be prone to fouling or proteolysis and should be cheap, portable and capable of being used by semi-skilled operators.

Protein engineering

Protein engineering is the process of developing useful or valuable proteins. It is a young discipline, with much research currently taking place into the understanding of protein folding and protein

recognition for protein design principles. There are two general strategies for protein engineering. The first is known as *rational design*, in which the scientist uses detailed knowledge of the structure and function of the protein to make the desired changes. This has the advantage of being generally inexpensive and easy, since site-directed mutagenesis techniques are well developed. Computational protein design algorithms seek to identify amino acid sequences that have low energies for target structures. While the sequence-conformation space that needs to be searched is large, the most challenging requirement for computational protein design is a fast, yet accurate, energy function that can distinguish optimal sequences from similar suboptimal ones. The second strategy is known as directed evolution. This is where random mutagenesis is applied to a protein, and a selection regime is used to pick out variants that have the desired qualities. Further rounds of mutation and selection are then applied. This method mimics natural evolution and generally produces superior results to rational design. An additional technique known as DNA shuffling mixes and matches pieces of successful variants in order to produce better results. This process mimics recombination that occurs naturally during sexual reproduction. Rational design and directed evolution techniques are not mutually exclusive; good researchers will often apply both. In the future, more detailed knowledge of protein structure and function, as well as advancements in high-throughput technology, will greatly expand the capabilities of protein engineering.

Nano-biotechnology

Nano-biotechnology is one of the most future-oriented technologies of the 21st century. Nano-biotechnology is becoming established in the interdisciplinary area, supported by a dynamic boom in innovations and will promote close cooperation between the life sciences, physical and chemical sciences as well as engineering. It is bridging the gap between inanimate and animate nature and aims at understanding the basic principles of biological functional units as well as creating extremely small components at “nano-scale” in a controlled way with technical materials. In April 2000, the BMBF established the “Nano-biotechnology” funding measure in order to develop this new research area in Germany and to make it internationally competitive. The convergence of nanotechnology and biomedical sciences opens a wide variety of biological research topics and medical uses at the molecular and cellular level. The long-term goal of nano-biotechnology is to contribute to the information transfer and to the explosive growth of this important research area. Nano-biotechnology used in biotechnology is definitely a life-saving innovation that has come as a boon to mankind. The real applications can turn around the life cycle of human beings in the years to come.

Molecular marker technology

Molecular marker technology has opened exciting new windows to view the natural biological world. DNA-based molecular markers have acted as versatile tools and have found their own position in various fields like taxonomy, physiology, embryology, genetic engineering, behavioral ecology, organism evolution, genetics, ecology, phylogeny, etc. DNA markers offer several advantages over traditional phenotypic markers, as they provide data that can be analyzed objectively. Molecular marker is any kind of molecule indicating the existence of a chemical or physical process. Molecular markers are used in molecular biology and biotechnology experiments to

identify a particular sequence of DNA. As the DNA sequences are highly specific, they can be identified with the help of the known molecular markers, which can find out a particular sequence of DNA from a group of unknown. Molecular markers have been used to monitor DNA sequence variation in and among the species and create new sources of genetic variation by introducing new and favorable traits from landraces and related species. In recent years, different marker systems such as Restriction Fragment Length Polymorphisms (RFLPs), Random Amplified Polymorphic DNAs (RAPDs), Sequence Tagged Sites (STSs), Amplified Fragment Length Polymorphisms (AFLPs), Simple Sequence Repeats (SSRs) or microsatellites, Single Nucleotide Polymorphisms (SNPs) and others have been developed and applied to plants and animals. Improvements in marker detection systems and the techniques used to identify markers linked to useful traits has enabled great advances to be made in recent years.

Marker-assisted selection technology

Marker assisted selection is a complementary technology for use in conjunction with more established conventional methods of genetic selection for plant and animal improvement. Marker assisted selection (MAS) involves establishing the linkage between the inheritance of a particular trait, which might be desirable as in the case of milk yield, or undesirable, as in the susceptibility to a disease, with the segregation of particular genetic markers. Thus, even if the gene that controls the trait is unknown, its presence can be inferred from the presence of the marker that segregates with it. MAS is an established plant breeding technology widely used by plant breeding programs around the world to supplement and enhance field-based phenotypic selection programs. Based on copious genetic and chromosome mapping research published in recent years, usable molecular markers that are associated with economic traits of interest are now available for public use. This technology is user-friendly and can be utilized at a relatively low cost by programs with limited technological capabilities. MAS can be used in an applied breeding program with two basic strategies. The first strategy recognizes that the superior genetic background identified by long term selection and evaluation is the most important aspect of an elite cultivar. Lines exhibiting statistically improved grain yield potential are rarely identified and highly valued when found since they likely represent a unique new combination of genes. Commonly during selection, lines possessing a superior combination of the key traits of grain yield, adaptation, and quality are identified, which lack other simply inherited traits that would enhance or protect their performance. Thus, this strategy attempts to improve and enhance genetic backgrounds documented to have superior yield performance by correcting deficiencies and/or enhancing the levels of secondary, simply inherited genetic traits that can be improved relatively easily using marker assisted selection. A second strategy for MAS is to cross two elite parents, one of which carries economically important genes that can be identified by appropriate molecular markers. Progeny are screened during generation advancement for molecular markers and progeny carrying markers associated with traits of value are retained. Selection for yield and other traits of importance is imposed secondary to selection for the gene being followed by MAS. Both types of MAS are valuable and necessary in an applied program for cultivar development.

Integrating MAS into breeding strategies could become increasingly important in the coming years to realize genetic gains with greater speed and precision. The promise of MAS for improving

polygenic traits in a quick time-frame and a cost-effective manner is still elusive. There is a wider appreciation that simply demonstrating that a complex trait can be dissected into QTLs and mapped to approximate genomic locations using DNA markers would not serve the ultimate goal of trait improvement. In facing the challenge of improving several lines for quantitative traits, MAS strategies use DNA markers in one key selection step to maximize their impact.

Plant transgenic technology

Transgenic technology has provided important tools for sustainable development of agriculture and forestry and can be of significant help in meeting the food needs of a growing and increasing urbanized population. The genetic engineering of plants by inserting a foreign DNA segment into the plant genome is called transgenesis and the developed plant a transgenic plant. All genetically engineered plants are GMOs (Genetically Modified Organisms), also known as LMOs (Living Modified Organisms). Common LMOs include agricultural crops genetically modified for agronomic traits. The advantage or potential benefit of transgenic plants is the applications for which they are being developed, the new traits being introduced. In brief, a GM plant allows expression of a trait not found inherently, regulated temporal and spatial expression of desired trait. With depleting natural resources and changing global climate, conventional agricultural practices alone are unable to sustain the quality and quantity of the produce. With the advent of modern biotechnology, newer tools permitting gene transfer across the species: transgenics, opened an avenue for solving the age old problem. Genetically modified crops can be produced by several ways, be it a physical or chemical delivery method of inserting gene of choice. The aims of gene transfer are – breeding crops with improved traits – as well as molecular farming, the production of pharmaceuticals, therapeutic molecules or specialty chemicals for industry. Transgenics have potential to reduce usage of insecticides, herbicides and weedicides, improve tolerance to stress, and function as edible vaccine and biosensors.

Bioinoculant technology

Bioinoculant technology deals with the prospect of improving agriculture by inoculation with biofertilizers, which is of great importance for less intensive, ecofriendly, restore natural soil fertility and low-input agricultural system existing in many developing countries. Agricultural fields that are deficient in a nutrient and are devoid of native populations of microorganisms capable of supplying the deficient nutrients are the appropriate locations for using biofertilizers for inoculations. The need for the use of biofertilizers has arisen primarily due to two reasons. First, because increase in the use of biofertilizers leads to increased crop productivity, and second because increased usage of chemical fertilizer leads to damage in soil texture and raises other environmental problems. Broadly, bioinoculants include biofertilizers, biopesticides and organic decomposers. Biofertilizers are live cells of beneficial microbial isolates that provide necessary nutrients (nitrogen, phosphorous, etc.), excrete growth promoting compounds and provide resistance to a variety of diseases that culminates to enhanced yield and production. While biopesticides are live microbial isolates or their metabolic products that eradicate/kill known insects/pests of crops. The third component of bioinoculants is the organic decomposers that include certain fungal species, bacterial genera and

actinomycetes that hasten decomposition of organic compounds and make available nutrients held as organic matter. The main types of bioinoculants, which are present in market include:

- (1) soil inoculants: to improve soil biological activity;
- (2) plant inoculants: to improve plant nutrient uptake and disease resistance; and
- (3) residue inoculants: to improve the breakdown of crop and litter residues.

Before using a bioinoculant product you need to be sure of the purpose that you want to add it for. Is it for soil or plant biology? You then need to match the product to your specific need.

Bioinoculant technology includes selection of appropriate bacterial strains, large scale production of their biomass, development of formulations (liquid, powder or granules), and their applications methods in the field. Reliable commercial use of bioinoculants on a large scale depends on formulations that are simple, economical, unaffected by long storage, and easy to transport and apply. It also depends on application technology that facilitates timely, easy, and precise delivery in the field. The study of a formulation process is generally undertaken when the choice of an efficient strain has already been made. However, the development of an economically viable production processes is a significant hurdle in the commercialization of bioinoculants. A primary method of achieving this goal is through strain engineering. Evolutionary engineering has been practiced for decades in the form of classic strain improvement. The process of genetic diversification and functional screening has now become a powerful means of improving the function of diverse biological systems from genes and enzymes to whole genomes. Gene shuffling is a method for effecting genetic diversification in evolutionary engineering programs that incorporate recombination into the evolutionary algorithm. This approach dramatically accelerates the process of directed evolution and is arguably the most robust method for the purposeful manipulation of biological structure and function. In addition, the optimization of formulation is largely independent of the strain, since strains within one genus or species share many cultural properties. A process developed for one strain of a genus may also be generally applicable to the other strains of the same genus, requiring only a few adaptations or modifications.

Embryo transfer technology

Embryo transfer technology is a technique by which embryos are collected from a donor female and transferred to recipient females, which serve as surrogate mothers for the remainder of pregnancy. Embryo transfer techniques have been applied to nearly every inter-generic and intra-specific species of domestic animals and too many species of wildlife and exotic animals, including humans and non-human primates. Within the past decade, the degree of sophistication of embryo transfer procedures has evolved to permit complete utilization of non-surgical procedures in the cow, long term embryo culture and storage with cryopreservation, and, more recently, micromanipulation and many techniques associated with genetic engineering. This is a rapidly developing science with a very short lag-time between discovery and application. The potential applications continue to increase with the development of new technology. This technology achieves a surprisingly rapid rate of multiplication of animals of the selected superior genotype. In natural course, a single female will produce a single progeny in about one year. But using super-ovulation and embryo transfer technology, it is possible to collect around 36 embryos from one female in one year. Assuming an average success rate of 50% in the embryo transfer, an average of 18 progeny

can be derived from one superior female in one year. Although the embryo transfer procedure has, fundamentally, remained unchanged, modifications in the technique to improve the efficiency, the donor and recipient safety, and the success rate, as well as the advent of embryo freezing, embryo micromanipulation and embryo sexing have kept the industry dynamic and extremely challenging.

Stem cell technology

Stem cells are characterized by the ability to renew themselves through mitotic cell division and differentiating into a diverse range of specialized cell types. Although most cells of the body, such as heart cells or skin cells, are committed to conduct a specific function, a stem cell is uncommitted and remains uncommitted, until it receives a signal to develop into a specialized cell. Research in the stem cell field grew out of findings by Canadian scientists Ernest A. McCulloch and James E. Till in the 1960s. The two broad types of mammalian stem cells are embryonic stem cells that are isolated from the inner cell mass of blastocysts, and adult stem cells that are found in adult tissues. Highly plastic adult stem cells from a variety of sources, including umbilical cord blood and bone marrow, are routinely used in medical therapies. Embryonic cell lines and autologous embryonic stem cells generated through therapeutic cloning have also been proposed as promising candidates for future therapies. Stem cells may hold the key to replacing cells lost in many devastating diseases like Parkinson's disease, diabetes, chronic heart disease, end-stage kidney disease, liver failure, and cancer are just a few for which stem cells have therapeutic potential. Today, science has brought us to a point where the immune response can be subdued, so that organs from one person can be used to replace the diseased organs and tissues of another. Scientists in academic and industrial research are vigorously pursuing all possible avenues of research, including ways to direct the specialization of adult and embryonic stem cells to become pancreatic islet-like cells that produce insulin and can be used to control blood glucose levels. Researchers have recently shown that human embryonic stem cells be directly differentiated into cells that produce insulin. Predicting the future of stem cell applications is impossible, particularly given the very early stage of the science of stem cell biology. To date, it is impossible to predict which stem cells those derived from the embryo, the fetus, or the adult or which methods for manipulating the cells, will best meet the needs of basic research and clinical applications. The answers clearly lie in conducting more research.

Animal transgenic technology

Animal transgenic technology deals with the production of transgenic animals that depends upon the success of a series of several complex technical experimental manipulations. Even under the best of circumstances, the success of these manipulations can be somewhat variable, depending upon the genetic background of the host or the recipient cells as well as upon the biological consequences of gene or DNA-sequence modification. Nevertheless, the construction of transgenic animals is available as a resource in most research settings, thus allowing for the systematic *in vivo* evaluation of the consequences of deleting, altering, or over-expressing a single gene or DNA sequence of interest. While both transgenic rats and mice useful for toxicological studies have been generated, thus far most efforts have focused on the generation of transgenic mouse model systems. Indeed, these systems provide excellent experimental models for the evaluation of gene-specific interactions in the studies of metabolism, pharmacology, and carcinogenesis.

Potentially, specifically modified transgenic animals will be generated that allow for a limited response to a toxic agent, thereby better defining the role of a specific gene product in the cellular or organism response pathway. In addition to the development of new and more elaborate transgenic models, one can expect to see an increasing number of applications of molecular biology in studies of toxicology in the future.

Liposome technology

Liposomes are amphipathic in nature, i.e., one part of their structure is water-soluble (hydrophilic or water-loving) and the other part is oil-like (hydrophobic or water-fearing or fat-soluble). They are well-recognized delivery vehicles that have been developed to enhance the therapeutic activity of several drugs. Liposomes are the smallest artificial vesicles of spherical shape that can be used as drug carriers and loaded with a great variety of molecules, such as vaccines, proteins, genes, enzymes and even plasmids to the body. Liposomes are extremely versatile and due to the variability of their composition they can be used for a large number of applications such as gene therapy, cancer therapy, clinical pathogenesis, enzyme replacement therapy, chelation therapy, immunomodulation, thermal transitions of phospholipids, neurobiological applications, leukocyte phagocytosis, botanical research, drug interactions with membranes, bacterial cell wall synthesis, lectin receptor interactions, microvillus formation in cells and many more. These are only a few of the applications for which liposomes have been used. Reports of new types of liposomes and as well as new uses for liposomes appear almost daily.

Vaccine technology

Vaccines are suspensions of live, killed, or attenuated microorganisms such as bacteria or viruses or their components. They are designed for human or veterinary use to prime the immune system to produce antibodies or cellular responses that ward off future diseases by pathogenic organisms or their toxins. Research into vaccine technology continues on many fronts to provide safe, efficacious, and affordable vaccine products. One approach to improve coverage that is being adopted by many vaccine manufacturers is to combine antigens into single preparation so that immunity to two or more diseases can be achieved with a single injection. Time release polymeric preparations are also being explored even for monovalent preparations to achieve repeated antigenic stimulation of the immune system without the need for multiple injections. Other research initiatives seek to eliminate the need for needles for delivery of vaccines through the use of oral vaccines, aerosolizers for inhalation, or needleless devices that use compressed gases at high pressure to deliver transdermal doses of vaccine preparations. The tremendous amount of research in the past few years has led to the development of vaccines against disease targets associated with bacterial, viral, and parasitic infections and provides hope for combating autoimmune and inflammatory diseases as well as cancer.

Synthetic biologicals technology

There is no agreed definition of synthetic biology, but it is best understood as the deliberate design of biological systems and living organisms using engineering principles. Synthetic biology is the

design and construction of new biological entities, such as enzymes, genetic circuits, and cells, or the redesign of existing biological systems. The goal of synthetic biology, which builds on advances in molecular, cellular, and systems biology, is to transform biology in the same way that synthesis transformed chemistry and integrated circuit design transformed computing. The aims of synthetic biology include:

- (1) the production of minimal living genomes;
- (2) the design of interchangeable parts that can be assembled into pathways for the fabrication of novel components;
- (3) the construction of entirely artificial cells; and
- (4) the creation of synthetic biomolecules.

Research within synthetic biology can be explored through one of the two approaches: top-down, or bottom-up. The top-down approach attempts to eliminate the problem of natural complexity by removing it, e.g., by stripping a genome of all genetic material that is not absolutely essential for replication and functionality. The bottom-up approach uses naturally occurring organisms that appear to have little complexity and adds the required functions by engineering them into the existing genome. There are a number of major areas of research that constitute the field of synthetic biology such as making minimal genomes; designing modular components; pathway engineering; expanding the genetic pool; production of artificial cells; and creation of synthetic biomolecules. Just as engineers now design integrated circuits based on the known physical properties of materials and then fabricate functioning circuits and entire processors (with relatively high reliability), synthetic biologists will soon design and build engineered biological systems.

Bioengineering

Bioengineering (also known as Biological Engineering) is the application of engineering principles to address challenges in the fields of biology and medicine. Bioengineering integrates physical, chemical, biochemical and mathematical competencies and expertise with engineering approaches for applications in the field of biology, medicine, health and food industry. It applies engineering principles to the full spectrum of living systems. This is achieved by utilizing the existing methodologies in such fields as molecular biology, biochemistry, microbiology, pharmacology, cytology, immunology and neuroscience and applies them to the design of medical devices, diagnostic equipment, biocompatible materials, and other important medical needs. Bioengineering is not limited to the medical field. Bioengineers have the ability to exploit new opportunities and solve problems within the domain of complex systems. They have a great understanding of living systems as complex systems, which can be applied to many fields including entrepreneurship. Much as other engineering disciplines also address human health (e.g., prosthetics in mechanical engineering), bioengineers can apply their expertise to other applications of engineering and biotechnology, including genetic modification of plants and microorganisms, bioprocess engineering, and biocatalysis. However, the main fields of bioengineering may be categorized as: Biomedical Engineering, Biomedical technology, Biomedical Diagnosis, Biomedical Therapy, Biomechanics, Biomaterials, Genetic Engineering, Cell Engineering and Tissue Culture Engineering. “Bioengineer-

ing” is also used to describe the use of vegetation in civil engineering constructions. The term bioengineering may also be applied to environmental modifications such as surface soil protection, slope stabilization, watercourse and shoreline protection, windbreaks, vegetation barriers including noise barriers and visual screens, and the ecological enhancement of an area.

Immobilized biocatalyst technology

Immobilized biocatalyst technology can be defined as “enzymes physically confined or localized in a certain defined region of space with retention of their catalytic activities, and which can be used repeatedly and continuously”. Accordingly, enzymes modified to a water-insoluble form by suitable techniques satisfy this definition of immobilized enzymes. This is known as immobilization and may be achieved by fixing the enzyme to, or within, some other material. It provides stability and easy separation of the enzyme from the products of the catalyzed reaction that minimizing downstream processing costs and possible effluent handling problems, particularly if the enzyme is noticeably toxic or antigenic. It also allows continuous processes to be practicable, with a considerable saving in enzyme, labor and overhead costs. During the last three decades, immobilization of enzymes, cells, and organelles has found vast analytical, technological, and medical applications. Biological materials have a great variety of properties, and for each particular application, the proper choice of support and immobilization method may not always be easy to accomplish. Immobilized biocatalyst technology is presently going through a phase of maturation and evolution. The evolution is shown by the ever broadening range of enzymic applications. Immobilized enzymes will clearly be more widely used in the future and this will be reflected in the number of enzymes available on an industrial (and research) scale, the variety of reactions catalyzed and the range of environmental conditions under which they will operate. Established immobilized enzymes will be put to new uses and novel enzymes, discovered within their biological niches or produced by design using enzyme engineering, will be used to catalyze hitherto unexploited reactions. This is just the start of the immobilized biocatalyst technology era.

Bioprocess engineering

Bioprocess engineering is the sub-discipline within biotechnology that is responsible for translating the discoveries of life science into practical products, processes, or systems that can serve the needs of society. Bioprocess engineers work at the frontiers of biological and engineering sciences to “Bring engineering to life” through the conversion of biological materials into other forms needed by mankind. Application areas commonly associated with bioprocess engineering include the production of biofuels, ethanol, amino acids and other organic acids, antibiotics, design and operation of fermentation systems, development of food processing systems, application and testing of product separation technologies, design of instrumentation to monitor and control biological processes, and many more. Like other engineering disciplines, bioprocess engineers are trained in the application of engineering sciences and problem solving techniques. What separates them from other engineers is their understanding of how every engineering field relates to living materials. With the rapid growth of biologically based technologies, bioprocess engineers are uniquely qualified to solve the problems of today and tomorrow. The demand for bioprocess

engineers continues to grow. They provide a bridge between the research lab and the economic, large-scale implementation of biotechnologies and food production systems.

Metagenomics technology

Metagenome technology (metagenomics) uses a culture-independent approach to collect, sort, categorize and identify sequences of microbes retrieved from environmental samples such as soil or water. The emerging field of metagenomics provides a new way of viewing the microbial world that will not only transform modern microbiology, but also may revolutionize understanding of the entire living world. Through metagenomics, scientists can apply the power of genomic analysis to entire communities of microbes, bypassing the need to isolate and culture individual community members. Metagenomics involves studying the genetic makeup of many microbes in an environment simultaneously, and makes accessible the many types of microbes that cannot be grown in the laboratory and therefore cannot be studied using the central tool of classical microbiology. Metagenomics has been applied to many environments, including the oceans, soils, thermal vents, hot springs, and the human mouth and gastrointestinal tract. Areas in which metagenomics offers particular value include: medicine, alternative energy, environmental remediation, biotechnology, agriculture, biodefense, and forensics. Increased understanding of microbes will lead to new concepts of genomes, species, evolution, and ecosystem robustness that will have effects beyond the specific field of microbiology. Metagenomics can help scientists address questions such as “How do microbes evolve?” “What is the role of microbes in maintaining the health of their hosts?” and “How diverse is life?”.

Bioremediation technology

Bioremediation is a process which relies on microorganisms to break down and detoxify organic chemicals such as hydrocarbons and some organochlorines. Carbon dioxide and water are the final degradation products for hydrocarbon wastes using this process. Bioremediation has a number of applications within the mining industry, including the treatment of the various types of wastes. Bioremediation has proven to be a non-disruptive, non-toxic, non-pathogenic cost-effective and highly efficient method of safely breaking down many environmentally persistent toxic chemicals. Bioremediation is gaining attention as it is fast, economical, and does not require heavy machinery or skilled labor, according to the analyst of this research service. These technologies are also efficient in the biodegradation of contaminants to create end products that are not hazardous to the environment. Bioremediation technologies in various companies generally vary from one site to another depending on the kind of contaminants, the hydrogeology, and the accessibility of the contaminants.

BIOTECHNOLOGY REGULATIONS

The National Institute of Health (NIH) was the first federal agency to assume regulatory responsibility in the United States. The Recombinant DNA Advisory Committee of the NIH published guidelines for working with recombinant DNA and recombinant organisms in the laboratory. Agencies responsible for the biotechnology regulations are: US Department of Agriculture (USDA)

regulates plant pests and medical preparation from living organisms, Environment Protection Agency (EPA) regulates pesticides and herbicides, and the Food and Drug Administration (FDA) ensures that the food and drug products are safe and effective.

The Regulatory framework in India: To ensure the compliance of safeguards on various levels, drafting of guidelines and their implementation is indispensable. The activities and use of GE organism and products thereof are regulated under the “Rules for the manufacture, use/import/export and storage of hazardous organisms/GE organism or cells notified under the Environment Protection Act, 1986, commonly referred as Rules, 1989. These rules and regulations are implemented by the Ministry of Environment and Forest (MOEF) & Department of Biotechnology (DBT) and state governments. Six competent authorities and their composition have been provided for in the Rules to handle various aspects (<http://www.dbtbiosafety.nic.in-stepwise-procedure>) are as follows:

- Recombinant DNA Advisory Committee (RDAC)
- Review Committee on Genetic Manipulation (RCGM)
- Genetic Engineering Approval Committee (GEAC)
- Institutional Bio-safety Committee (IBSC)
- State Border Coordination Committee (SBCC)
- District Level Committees (DLCs)

While RDAC is advisory in function, the IBSC, RCGM & GEAC are for regulatory functions, SBCC & DLC are for monitoring purpose. In addition to the above, a monitoring-cum-evaluation committee (MEC) has been set up by RCGM to monitor the field performance of GM crops. DBT has issued certain guidelines for applicants seeking approval for environmental release of GE plants in India under Rules, 1989. Specifically these protocols address key elements of the safety assessment of foods and/or livestock feeds that may be derived from GM crops. The results are submitted to appropriate regulatory bodies, i.e., RCGM and GEAC as required. DBT reviewed these guidelines in 2008 and made them available as draft guidelines on its website. DBT has prepared five protocols each based on best international practices including guidance and peer reviewed from the Codex Alimentarius Commission, FAO, WHO and Organization for Economic Cooperation and Development and the ILSI (<http://www.dbtbiosafety.nic.in-protocols>). These include:

- Acute oral safety limit study in rats or mice
- Sub-chronic feeding study rodents
- Protein thermal stability
- Pepsin digestibility assay
- Livestock feeding assay

For additional guidance, the ICMR guidelines for the safety assessment of foods derived from genetically engineered plants that provide a comprehensive summary of information and data required must be provided to regulatory authorities of demonstrate the GE plants, may also be reviewed (<http://www.dbtbiosafety.nic.in-protocols>). With the increase in Indian biotechnology sector, significant implementation in area of regulation is expected. The 2004 Report of Task Force on Application of Agricultural Biotechnology (The Swaminathan Report) and the 2005 report of

Task Force of Recombinant Pharma (The Mashelkar report) both have recommended the establishment of a national biotechnology regulatory authority (NBRA). The NBRA will be an autonomous, statutory agency established by GOI with a mandate to safeguard the health of the people of India and to protect the environment by identifying risks posed by, or as a result of, modern biotechnology, and deployment of biotechnology products and processes. In order to empower NBRA, DBT has defined a new legislation namely the Biotechnology Regulatory Bill in April 2008 (www/agbios.com/sabp – June 2008).

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